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Original Report

Subjective and objective quantification of physician's workload and performance during radiation therapy planning tasks

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Abstract

Purpose: To quantify, and compare, workload for several common physician-based treatment planning tasks using objective and subjective measures of workload. To assess the relationship between workload and performance to define workload levels where performance could be expected to decline.

Methods and Materials: Nine physicians performed the same 3 tasks on each of 2 cases ("easy" vs "hard"). Workload was assessed objectively throughout the tasks (via monitoring of pupil size and blink rate), and subjectively at the end of each case (via National Aeronautics and Space Administration Task Load Index; NASA-TLX). NASA-TLX assesses the 6 dimensions (mental, physical, and temporal demands, frustration, effort, and performance); scores > or ≈ 50 are associated with reduced performance in other industries. Performance was measured using participants' stated willingness to approve the treatment plan. Differences in subjective and objective workload between cases, tasks, and experience were assessed using analysis of variance (ANOVA). The correlation between subjective and objective workload measures were assessed via the Pearson correlation test. The relationships between workload and performance measures were assessed using the t test.

Results: Eighteen case-wise and 54 task-wise assessments were obtained. Subjective NASA-TLX scores ($P < .001$), but not time-weighted averages of objective scores ($P > .1$), were significantly lower for the easy vs hard case. Most correlations between the subjective and objective measures were not significant, except between average blink rate and NASA-TLX scores ($r = -0.34$, $P = .02$), for task-wise assessments. Performance appeared to decline at NASA-TLX scores of ≥ 55.

Conclusions: The NASA-TLX may provide a reasonable method to quantify subjective workload for broad activities, and objective physiologic eye-based measures may be useful to monitor workload for more granular tasks within activities. The subjective and objective measures, as herein quantified, do not necessarily track each other, and more work is needed to assess their utilities. From a series of controlled experiments, we found that performance appears to decline at subjective workload levels ≥ 55 (as measured via NASA-TLX), which is consistent with findings from other industries.

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Introduction

Workload is widely recognized as an important factor contributing to suboptimal human performance and errors.[1] Workload is a hypothetical construct that represents the overall cost incurred by a human operator to achieve a particular level of performance.[2] Indeed, some industries have performed extensive research to understand, measure, and define safe workload levels to ensure the optimal system design and cognitive control of operators.[3,4]

Within radiation oncology, there has been an increased focus on safety,[1,5,6] including a recent study to quantify workload among radiation therapy (RT) professionals.[7] Mazur et al[7] used the National Aeronautics and Space Administration Task Load Index (NASA-TLX) instrument in real clinical environment to subjectively measure workload levels among 21 RT professionals and noted potentially unsafe work levels for some tasks (eg, global NASA-TLX scores > or ≈ 50, the upper limit of workload as defined by standards used in other industries).[8,9] NASA-TLX is based on a multidimensional rating procedure that considers 6 dimensions (mental, physical, and temporal demands, frustration, effort, and performance) to yield a global workload score between zero and 100.

This comparison to standards from other industries was necessary as workload standards within radiation oncology, or in medicine more broadly, have not generally considered the NASA-TLX construct but rather have mostly focused on staffing levels, work times, and resource requirements.[10–12]

Shortcomings of the study by Mazur et al[7] include their reliance on subjective workload measures and lack of performance data. To address these shortcomings, we investigated physician's workload during RT planning tasks using subjective measures (via NASA-TLX), and objective measures (via pupil diameter and blink rate). This latter objective approach has been successfully used in a variety of industries including aviation, transportation, nuclear power, and medicine.[13-15] Overall, experts suggest that increased workload generally increases pupil diameter and reduces blink rate.[16,17] Our primary aim was to quantify the workload for several common physician-based treatment planning tasks using both objective and subjective measures. Second, we explored the correlation between these subjective and objective measures. Third, we explored the association between workload and performance, to identify workload levels where performance degradation could be expected.

Methods and materials

Broad overview of participating staff

Nine physicians (4 experienced faculty and 5 residents) volunteered to participate in this Institutional Review Board (IRB)-approved study. All participants were oriented to the experimental goals and residents were given a $100 gift card for participating.

Broad overview of simulated environment and design of experiments

The experiments were conducted in the "human factors workroom," designed to resemble the routine clinical environment (Fig 1). Participants performed the usual tasks for treatment planning for 2 cases; case 1 was relatively "simple" (palliative opposed lateral 2-field brain); case 2 was relatively "complex" (curative 4-field postoperative pancreas). For each case, physicians performed 3 successive tasks (see Table 1 for detailed step-by-step procedures). Participants were told to complete case 1 and case 2 within 10 and 15 minutes, respectively; however, they were permitted to complete the cases even if they extended their work beyond the allocated time. A large digital clock timer, placed immediately adjacent to the computer monitor, counted down the allotted time

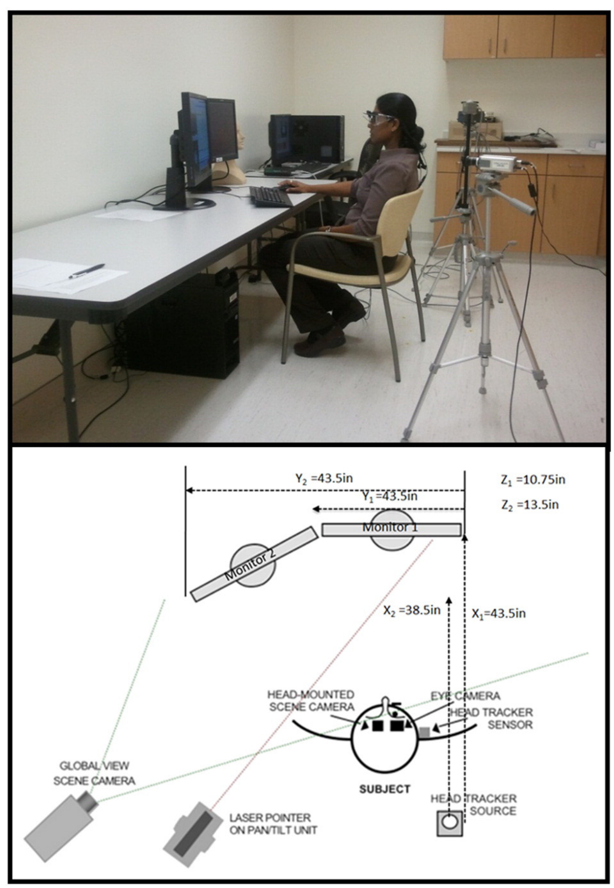

Figure 1 Illustrations of the study conditions. The subjects were placed in front of computer monitors in a manner intended to replicate typical clinical settings. (Top) Photograph of the human factors workroom. (Bottom) Graphical representation (top view) of the layout during experiments. The layout is largely similar to that seen in many clinics.

Table 1 Step-by–step procedures

Procedure
Access patient record in electronic medical record (*EMR)
Task 1 Start
1. Review patient's written reports to familiarize oneself with the case, including records in the departmental EMR, in the clinic notes, and in the radiology reports.
2. Type a note into the departmental EMR to document the planned simulation, including instructions to the simulator for the pending CT, and to dosimetry for the anticipated doses (done in the notes section and Quick Orders section of the EMR, respectively).
Task 1 End
Go to treatment planning system to retrieve the plan
Task 2 Start
3. Review diagnostic images within the planning system.
4. Segment the CT image to define the target volume (if desired; not needed for case 1, but was needed for case 2). Review contours/segmentations generated by the dosimetrist (of the normal anatomy).
5. Design your treatment field(s).
a. Create individual beams.
b. Save plan.
Task 2 End
Simulation and dosimetry is done instantaneous (in background)
Task 3 Start
6. Review the generated plan.
a. Retrieve plan.
b. Review Isodose distribution.
c. Review dose volume histograms, if needed.
d. Review spreadsheet.
7. Approve plan if acceptable. You might decide not to approve it.
Go back to EMR
8. Review and approve (if agreed) the prescription.
Task 3 End

*MOSAIQ, Version 2.1, Elekta AB, Stockholm, Sweden.

remaining to complete the case, so the participants were aware of the passage of time.

Participants wore a head-mounted eye-tracking device (VisionTrak, ISCAN, Inc, Burlington, MA) during the experiments to collect data on blink rate and pupil diameter. The wearing of the goggles could theoretically influence physician performance (and patient safety) and thus was not done in the clinical setting.

Workload measures (NASA-TLX and physiological eye characteristic data)

Following each laboratory experiment, participants completed a subjective workload assessment of the entire case and the 3 individual tasks, using the broadly used and validated NASA-TLX instrument.[2] Investigators were available to assist participants during this scoring to answer questions and provide clarification as to how to complete the instrument.

The objective workload assessments were collected via pupil diameter and blinks throughout the cases at a 60-Hz sampling rate while allowing complete freedom of head movement. The average pupil diameter and average blink rate were calculated using a time-weighted average for each task for each case. Pupil diameter was estimated using the average of horizontal and vertical number of pixels captured by the eye-tracking video system. Eye blinks were counted if the eye closure was ≥80%.

The subjective and objective workload measures were summarized using descriptive statistics, with comparisons made using analysis of variance (ANOVA). Patterns (NASA-TLX scores, average pupil diameter, and average blink rate score transitions between tasks in sequence) were assessed using a paired *t* test. Data from faculty versus residents were compared using a *t* test.

Correlation between subjective and objective workload measures

Correlations between the subjective workload measure (NASA-TLX scores) and each objective workload measure (average pupil diameter and average blink rate) were assessed using the Pearson correlation coefficient.

Performance measure

At the end of each case, the participants subjectively commented about their overall performance and also stated their willingness to approve the plan that they created for treatment using a dichotomous "Yes" versus "No."

Relationship between workload and performance

For each case, each subjective and objective workload measure (NASA-TLX, average blink rate, and average pupil diameter) was compared with the willingness to approve the plan (Yes vs No) via a *t* test.

Upper workload limit ("redline")

In other fields, researchers have found that the transition from a moderate to a high level of workload is often considered as the upper workload limit (or workload "redline").[18] Thus, we constructed the receiver operating characteristic (ROC) curve(s) to evaluate the predictive ability (eg, sensitivity vs 1-specificity) of subjective and objective workload measures for the endpoint of physician willingness to approve the treatment plan. The workload score with best predictive ability (a point on the ROC curve closest in linear distance to the upper left corner, or coordinate (0,1), of the ROC space) was taken as the potential "best" cutoff point (upper workload limit or redline).

Results

A total of 18 case-wise and 54 task-wise task assessments were made with 9 participants over a 45-day period. The estimated total time of data collection (including calibration procedures and training) in the laboratory was ≈21 hours.

Subjective workload measure (NASA-TLX)

Workload levels for case 1 were relatively moderate (NASA-TLX: overall average 49; tasks 1 to 3 average scores range from 40 to 45), and were higher for case 2 (NASA-TLX: overall average 66; tasks 1 to 3 average scores range from 59 to 66); $P < .001$ via ANOVA analysis on a case-wise basis (using the *t* test we also found statistical differences between individual tasks for case 1 vs individual tasks for case 2 [ie, case 1 task 1 vs case 2 task 1, etc.]). Within each case, ANOVA analysis found no significant differences between the tasks (eg, task 1 vs task 2 vs task 3) ($P > .1$). The paired *t* test within each case revealed no significant patterns (all $P > .1$). Residents displayed significantly higher NASA-TLX scores than faculty for the task-wise pooled data for both cases ($P = .03$), and task-wise data pooled for case 2 ($P = .02$), but not task-wise data pooled for case 1. No other significant differences in experience level were found (Fig 2).

Objective workload measure (physiological eye characteristic data)

Overall, both cases revealed relatively similar ranges of average pupil diameters (Fig 3); ANOVA analysis revealed no significant difference between the cases or tasks ($P > .1$). However, the paired *t* test within each case revealed the following patterns: increase of pupil diameter from task 1 to task 2 (case 1) ($P = .02$); increase of pupil diameter from task 1 to task 2 (case 2) ($P = .02$); and decrease of pupil diameter from task 2 to task 3 (case 1) ($P = 0.01$). The literature suggests that pupils dilate during more difficult tasks, suggesting the highest workload for task 2 in both cases.[14,15]

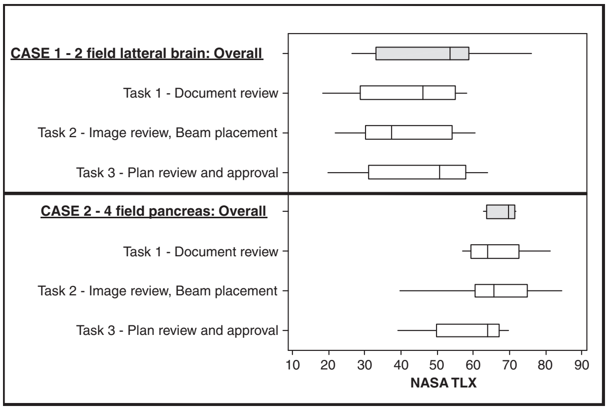

Figure 2 Box plot of average National Aeronautics and Space Administration Task Load Index (NASA-TLX) scores for each case and task.

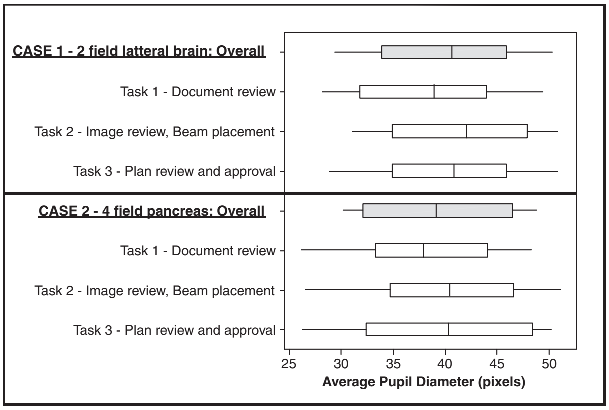

Figure 3 Box plot of average pupil diameters for each case and task.

Of the 54 tasks assessed, the blink rate data for 6 tasks were discarded due to unintended shifts of equipment during data acquisition, leaving 48 evaluable data sets. Overall, both cases revealed relatively similar ranges of average blink rates (Fig 4); ANOVA analysis revealed no significant difference between the cases or tasks. Interestingly, the paired *t* test within each case suggested a significant decrease of average blink rate from task 1 to task 2 for both experiments (case 1: $P = .01$; and case 2: $P = .09$). The literature suggests that blink rate reduces in more difficult tasks, which suggests highest workload for task 2 in both cases.[14,15]

There were no differences between the experienced faculty and residents in either the average pupil diameter or average blink rate measurements ($P > .1$).

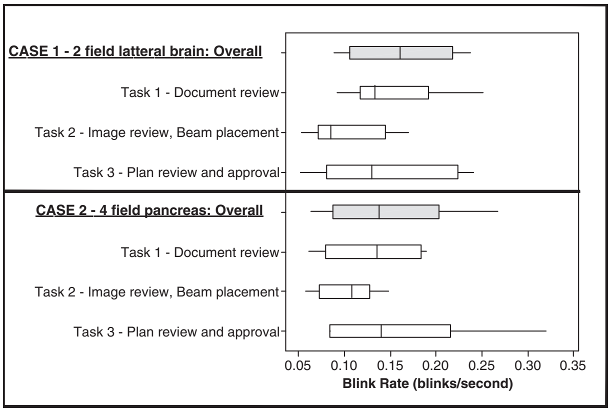

Figure 4 Box plot of average blink rates for each case and task.

Correlation between subjective and objective workload measures

There was a significant negative correlation between the average blink rate and NASA-TLX scores ($r = -0.34$, $P = .02$), for task-wise data pooled from 2 cases. The negative r value is as expected; a lower blink rate suggests a greater workload. All other correlations were not significant.

Performance

Five of the 18 cases were approved for treatment (case 1: 3 faculty and 1 resident approved, and for case 2: 1 faculty approved). The reasons for not approving treatment plan varied and included such things as a need for more information, experience limitations, or knowledge gaps.

Relationship between workload and performance

There were significantly lower NASA-TLX scores for participants who were willing to approve the plan (versus those not willing) (P = .004; Fig 5). This trend persisted when the data from the faculty and residents were considered separately. No significant relationships between objective workload measures and performance were detected.

Upper workload limit ("redline")

Because the NASA-TLX was the only workload measure related to the performance measure (ie, willingness to approve the plan), the analysis of a potential workload redline was conducted based on the NASA-TLX scores. The NASA-TLX score of 55 had the "best" predictive ability (sensitivity = 0.8, specificity = 1; point on the ROC curve closest to the upper left-hand corner).

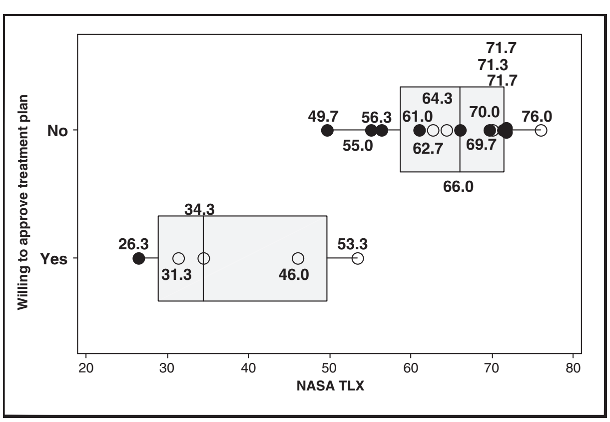

Figure 5 Scatter-plot of National Aeronautics and Space Administration Task Load Index (NASA-TLX) scores for approved versus not approved treatment plans. Open symbols represent the scores from the faculty, and closed symbols represent scores from the residents.

Discussion

Aim No.1: Physician's workload for "simple" versus "complex" cases

Our data suggest that physician's subjective assessment of workload is higher for the tasks in the 4-field pancreas case than for the opposed 2-field lateral brain case, which is consistent with the general perceptions regarding the efforts needed of a "simple palliative" versus a "complex curative" case. This supports the potential use of NASA-TLX as a means to quantify the differences in workload for such cases. Conversely, we did not note similar differences in the objective measures of average pupil diameter and average blink rate between the 2 cases, suggesting that the averages of these physiologic metrics may not be ideal metrics to quantify workload for these broad tasks. Nevertheless, within each case we found significant patterns for the manner in which these physiologic measures changed between tasks; task 2 typically having larger pupil diameters and lesser blink rates, relative to tasks 1 and 3. This observation suggests that these physiologic measures can be useful in identifying differences in workload, for example between specific sub-tasks. Further, it highlights the shortcomings of using a simple time-weighted average as a measure to describe the overall workload for the entire task or case as it assumes that pupil diameter and eye blinks are equally weighted throughout the task, which is not necessarily correct. For example, pupil diameter is known to quickly adapt to alterations in the luminance and subject's gaze angle within the visual field while performing a visual task.[16,17] However, as this was our first attempt to objectively measure workload within radiation oncology it was a justifiable approach for this study.

Aim No. 2: Subjective versus objective workload measures

For task-wise data pooled from both cases, there was a correlation between the average blink rate and the NASA-TLX scores. However, no correlation was found between the average pupil diameter and the NASA-TLX scores. The literature is also not uniform in this regard and presents examples where physiological eye characterizes data either do or do not respond to changes in perceived workload as expected.[13,14,19,20] In concert, our data suggest that average blink rate might be a reasonable measure of workload for some radiation therapy tasks performed continuously without interruption. Additional work is needed to better understand the potential utility of these physiologic measures.

Aim No. 3: Workload vs performance

The physician's willingness to approve a treatment plan declined with NASA-TLX scores > or ≈55, suggesting this as a potential upper workload limit. This is an exciting observation. In other industries (eg, aviation, etc.), the NASA-TLX score has been adopted as a meaningful measure of workload, and a NASA-TLX score > or ≈50 has been associated with reduced performance.[8,9] If this holds true in radiation oncology (or even more broadly, medicine), the NASA-TLX might provide a relatively straightforward way to identify tasks at risk of reduced human performance.

Conclusions

There are several limitations to this study, and thus caution should be exercised in generalizing our findings. First, the results are based on a study with a limited number of cases and participants from 1 department, performed on the set of specific tasks and software in a simulated laboratory environment, which does not comprehensively reproduce a real clinical setting. One could argue that many of the contributors to workload in the real clinic setting (eg, noise, interruptions, and the need to multitask) are absent in the laboratory, and thus our laboratory environment is artificial. In this regard, this bias would have been to underestimate the workload and overestimate performance. The laboratory findings nevertheless appear to complement findings from a clinic, where NASA-TLX scores for approved plans were <50.[7] Interestingly, in the laboratory setting we were able to create conditions where the workload was often higher (NASA TLX ≈ 26–76) than in the clinical setting reported by Mazur et al[7] (NASA TLX ≈ 25–51). Thus, as one might expect, the physician's performance in the laboratory was suboptimal (ie, they were sometimes not willing to approve the plan) while subjected to a high workload. We suppose that when physicians are presented with workload levels >55 in real clinical settings they (consciously or not) make modifications (ie, take more time to make decisions; review additional patient records; recheck their work; consult with other physicians, etc.) to assure the desired level of performance is achieved. The laboratory setting used in this report enabled us to perform the subjective and objective assessments. The objective eye movement data are challenging to assess in the clinical setting because the goggles worn to gather the data can be cumbersome, influence the physician performance, and potentially effect patient safety. Further, the laboratory setting allowed us to use the same clinical case for multiple physicians, thereby reducing potential variations stemming from inter-case differences. In the future, one might be able to obtain similar eye data in the clinical environment via non-obtrusive eye-tracking technology.

Second, we did not consider a full breadth of cases (eg, intensity modulation was not considered). However, we did perform a large number of assessments involving ≈21 hours of direct data collection. Further, a modest number of cases and subjects is typical for this type of research requiring extensive in-depth human assessment.

Third, the NASA-TLX might not be ideal for subjective assessments of physician's workload. Nevertheless, it is one of the most well accepted instrument to perform subjective workload assessments.

Fourth, pupil diameter and blink rate could be influenced by parasympathetic and sympathetic systems, and daytime and psychotropic substances (eg, coffee) that were not controlled for in our experiments. However, the consistency in the findings among participants suggests that the impact of these environmental factors was modest.

Fifth, we did not randomly assign the order of the cases to participants (ie, case 1 was always conducted first). This was done to control for potential learning effect between cases, including the familiarity with tasks, experimental setting, and equipment (ie, wearing goggles). The observation that the NASA-TLX defined workload in the second case was higher than the first suggests that this is a real finding, as any training effect would have tended to lessen the workload of the second case.

Sixth, we collected the NASA-TLX after the entire case, not after each task, in order to not interrupt participants during experiments. This could influence the scoring of pair-wise comparisons and specific dimensions of the NASA-TLX. Perhaps this caused the NASA-TLX scores within the tasks (for each case) to be relatively consistent as participants may have had difficulty separating "from their memory" the different tasks after completing the entire case.

Directions for future research include the following: (1) a larger sample size with a broader number or type of cases (eg, including intensity modulated radiation therapy), perhaps representing participants from different clinics and different software programs; (2) evaluation of additional subjective and objective measures of workload and performance in clinical and laboratory settings; and (3) evaluation of alternative means to extract quantitative metrics from the objective eye data beyond simple time-weighted averages.

Despite the limitations, this study is a reasonable first step toward understanding the use of subjective and objective measures of workload during treatment planning tasks in radiation oncology. The safe delivery of radiation therapy requires extreme mental concentration and expertise in a complex environment (eg, multiple electronic medical records systems and computer interfaces) and care should be taken to assure that workload demands do not exceed worker capabilities. Mazur et al[7] noted workload levels for some radiation oncology workers might exceed “safe” levels based on standards from other industries.

Pending the results from such studies, it might be reasonable to consider workload levels as an independent quality measure to assess the quality assurance of processes used to plan and deliver radiation therapy. There is a strong need to conduct additional workload studies for other cases and tasks to determine generalizable workload redlines for radiation therapy based on subjective and objective instruments and measures, to help radiation oncology clinics maintain safe and reliable systems.

Acknowledgments

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